

JUNHOO LEE

Ph.D. Candidate, Seoul National University

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EDUCATION

Seoul National University

Ph.D. Candidate in Intelligence and Information

Seoul, South Korea

Sep 2021 – Present (Exp. Aug 2026)

Seoul National University

B.Sc. in Electrical and Computer Engineering

Seoul, South Korea

Mar 2017 – Sep 2021

RESEARCH INTERESTS

I am interested in foundation models, especially in the scaling behavior of learned representations: what internal structure emerges as pretraining scales, what remains stable across model and data scale, and how these structures shape downstream behavior, adaptation, and controllability. My work connects model design, pretraining, representation learning, evaluation, and inference-time control across language, vision, generative, and multimodal settings. More broadly, I aim to understand not only how models become stronger with scale, but also what structure makes those gains possible and reliable.

SELECTED PUBLICATIONS

- **Unlocking the Potential of Diffusion Language Models through Template Infilling, ACL 2026** #Large Diffusion Language Model

Junhoo Lee, Seungyeon Kim, Nojun Kwak

Introduced Template Infilling and Dynamic Segment Allocation, an inference-time scaffolding framework for diffusion language models that improves reasoning, code generation, and planning.

- **CSF: Black-box Fingerprinting via Compositional Semantics for Text-to-Image Models, CVPR 2026** #Diffusion Model #Fingerprinting

Junhoo Lee, Mijin Koo, Nojun Kwak

Developed a query-only black-box attribution framework for fine-tuned text-to-image models, combining compositional semantic probing with Bayesian lineage inference to identify model family and origin across diverse fine-tuned variants.

- **What’s Making That Sound Right Now? Video-centric Audio-Visual Localization, ICCV 2025** #Audio-Visual Localization

Hahyeon Choi, Junhoo Lee, Nojun Kwak

Presented AVATAR, a video-centric audio-visual localization benchmark, together with TAVLO, a temporally aware model designed for realistic grounding scenarios involving mixed sounds, multiple entities, off-screen sources, and cross-event mismatches.

- **Deep Support Vectors, NeurIPS 2024** #Deep Learning Theory #Interpretability

Junhoo Lee, Hyunho Lee, Kyomin Hwang, Nojun Kwak

Introduced DeepKKT, a method for recovering support-like learned structures from pretrained classifiers, enabling explanation, practical dataset distillation, model inversion, and latent-space generative editing without access to the original training data.

- **Do Not Think About Pink Elephant! (Co-first), CVPR Workshops 2024** #Diffusion Model #Safety

Kyomin Hwang[†], Suyoung Kim[†], Junhoo Lee[†], Nojun Kwak

Studied how large generative models fail to represent negation, revealing a “white bear phenomenon” in which suppressed concepts remain active, and proposed prompt-based attack and defense strategies grounded in representation-space analysis.

- **Any-Way Meta Learning, AAAI 2024** #Meta Learning

Junhoo Lee, Yearim Kim, Hyunho Lee, Nojun Kwak

Improved episodic meta-learning by relaxing the fixed N-way setting and exploiting label equivalence across tasks.

- **SHOT: Suppressing the Hessian along the Optimization Trajectory, NeurIPS 2023** #Meta Learning

Junhoo Lee, Jayeon Yoo, Nojun Kwak

Studied how flattening the optimization trajectory improves adaptation behavior in gradient-descent-based meta-learning.

AWARDS & HONORS

BK21 Future Innovation Talent Bronze Prize (USD 1,000)	2023
BK21 Outstanding Research Talent Fellowship (USD 3,500)	2023
Yulchon AI Star Scholarship (USD 8,000)	2022
3rd Place, SNU FastMRI Challenge (USD 3,000)	2021
Kwanak Scholarship (full funded)	2021
National Science and Engineering Scholarship (full funded)	2017–2018

PUBLICATIONS

- **Unlocking the Potential of Diffusion Language Models through Template Infilling**
Junhoo Lee, Seungyeon Kim, Nojun Kwak
Annual Meeting of the Association for Computational Linguistics (ACL), 2026. (long paper, oral presentation).
- **CSF: Black-box Fingerprinting via Compositional Semantics for Text-to-Image Models**
Junhoo Lee, Mijin Koo, Nojun Kwak
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), June 2026.
- **End-to-End Multi-Entity Customization**
Wonhark Park*, Jaehyun Lee*, Wonsik Shin, Junhoo Lee, Nojun Kwak
IET Image Processing, Accepted. 2026.
- **Deep Edge Filter** (Co-first)
Dongkwan Lee[†], Junhoo Lee[†], Nojun Kwak
The Conference on Neural Information Processing Systems (NeurIPS), 2025.
- **What's Making That Sound Right Now? Video-centric Audio-Visual Localization**
Hahyeon Choi, Junhoo Lee, Nojun Kwak
The IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
- **Deep Support Vectors**
Junhoo Lee, Hyunho Lee, Kyomin Hwang, Nojun Kwak
The Conference on Neural Information Processing Systems (NeurIPS), 2024.
- **Any-Way Meta Learning**
Junhoo Lee, Yearim Kim, Hyunho Lee, Nojun Kwak
The AAAI Conference on Artificial Intelligence (AAAI), 2024.
- **SHOT: Suppressing the Hessian along the Optimization Trajectory**
Junhoo Lee, Jayeon Yoo, Nojun Kwak
The Conference on Neural Information Processing Systems (NeurIPS), 2023.
- **Do Not Think About Pink Elephant!** (Co-first)
Kyomin Hwang[†], Suyoung Kim[†], Junhoo Lee[†], Nojun Kwak
The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR Workshops) – Responsible Generative AI, 2024.
- **Coreset Selection for Object Detection**
Hojun Lee, Suyoung Kim, Junhoo Lee, Jaeyoung Yoo, Nojun Kwak
The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR Workshops) – Dataset Distillation, 2024.
- **Practical Dataset Distillation Based on Deep Support Vectors**
Hyunho Lee, Junhoo Lee, Nojun Kwak
The European Conference on Computer Vision Workshops (ECCV Workshops) – Dataset Distillation Challenge, 2024.
- **The Role of Teacher Calibration in Knowledge Distillation**
Suyoung Kim, Seonguk Park, Junhoo Lee, Nojun Kwak
IEEE Access, 2025.

RESEARCH PROJECTS

Research on Gradient-Descent-Based Meta-Learning Algorithms <i>Funded by Samsung Electronics Main Researcher</i> • Conducted research to uncover the mechanism behind the success of gradient-descent-based meta-learning algorithms and developed an improved method, resulting in a publication at a top-tier machine learning conference.	Sep. 2022 – Sep. 2023
Development of Self-Directed Visual Object Detection for Shopping Assistance <i>Supported by IITP (Institute of Information & Communications Technology Planning & Evaluation) Project Manager</i> • Conducted research on a shopping-oriented visual assistant that detects objects in user-submitted images, extracts product attributes, and continually acquires new object categories through OCR-based labeling and incremental learning.	Mar. 2022 – Nov. 2024
Development of AI Video Generation for Korean Traditional Media Art <i>Supported by MCST (Ministry of Culture, Sports and Tourism), in collaboration with EASYWITH Main Engineer</i> • Led the development of a Korean-style AI video generation system using curated domain-specific datasets of traditional media art; designed a data-centric pipeline for high-resolution (up to 8K) video generation, including dataset construction, content similarity evaluation, and automated production workflows.	Jul. 2024 – Mar. 2026
Physically Plausible Video Diffusion Modeling <i>Project under AI Star Fellowship Program (MSIT, Korea) Project Manager</i> • Led the development of video diffusion models that generate physically consistent dynamics and designed physics-aware modeling strategies to improve temporal coherence and realism in generated videos.	Mar. 2025 – Nov. 2025

SKILLS

Programming Languages: Python, CUDA, R, C++, Verilog

ML Development Stack: PyTorch, TensorFlow, Slurm, Docker

TALKS

Compositional Semantic Fingerprinting (CSF) for Black-box Attribution of Text-to-Image Models

Apr. 2026

SWCS 2026 (Software Convergence Symposium), KIISE

Deep Support Vectors: Interpreting Deep Models via Support Vector Extraction

Jun. 26, 2024

Korea Computer Congress (KCC 2024)

ACADEMIC SERVICE

Conference Reviewer: CVPR (2025, 2026), ICCV (2025), ECCV (2026), ICLR (2025, 2026), ICML (2026), NeurIPS (2024, 2025, 2026)

PATENTS

- **Method and Apparatus for Edge Filtering to Improve the Robustness of Deep Learning Models**, Korean Patent, 10-2025-0215458
- **Method and Apparatus for Controlling Unintended Object Generation in Image Generation Models through Prompt Reconfiguration**, Korean Patent, 10-2025-0063788